Deep_Tech/WESOnline MENTORSHIP PROGRAM

Project Executive Summary

End-of-Cohort Project Presentations

Project Title: P001-MVB: Explainable and Responsible AI for Differentiating Bacterial and Viral Meningitis – A Clinical Decision Support System	**Team Members:** Abubakar Amidu (C3 Data Science AI/ML 002) **Contact Email / Phone:** abubakaramidu@gmail.com / 08022951530
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How to Complete This Report: Write 40–50 words in each section. Be clear, specific, and avoid jargon. Each section answers a key question that a judge or evaluator would ask about your project. Fill in all sections before your presentation.

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Overview	P001-MVB is an Explainable and Responsible AI clinical decision-support system that differentiates bacterial from viral meningitis using cerebrospinal fluid and blood laboratory data. Designed for healthcare professionals, it combines machine learning, SHAP explainability, and Responsible AI governance to support transparent, evidence-informed clinical decision-making while maintaining mandatory human oversight.
Problem	Rapidly distinguishing bacterial from viral meningitis is clinically challenging, yet delayed diagnosis of bacterial cases can have serious consequences. Many AI models prioritise predictive accuracy without sufficient transparency, governance, or human oversight, limiting clinician trust and reducing confidence in deploying AI safely within real-world healthcare environments.
Solution / Product	P001-MVB combines a Logistic Regression classifier with SHAP-based explanations, a comprehensive Responsible AI governance framework, and a Pre-Decision Verification Checkpoint (PDVC) that identifies higher-risk predictions requiring clinician review. This approach prioritises patient safety, transparency, accountability, and responsible deployment rather than predictive performance alone.
How It Works	A clinician enters patient laboratory values into the Streamlit application. The system predicts bacterial or viral meningitis, displays confidence scores and SHAP explanations, evaluates governance triggers, and activates the Pre-Decision Verification Checkpoint whenever additional clinician verification is recommended before any clinical decision is made.
Impact Overview	P001-MVB demonstrates how Explainable and Responsible AI can support clinical decision-making through transparency and structured human oversight.

	The deployed prototype provides an educational resource for healthcare professionals, researchers, and students while showcasing practical methods for reducing automation bias. Future success will focus on validation using representative clinical datasets.
Industry Relevance	This project supports the healthcare and digital health sectors, where demand for trustworthy clinical AI continues to grow. As organisations increasingly adopt AI-assisted decision support, explainability, governance, and responsible deployment have become essential requirements for clinicians, regulators, researchers, healthcare institutions, and technology developers.
Next Steps	Following the mentorship programme, the project will continue through external validation using clinically representative datasets, prospective evaluation of the governance framework, improved input validation, and enhanced audit capabilities. The Responsible AI framework will also be extended to additional high-impact healthcare decision-support applications.